

# Lecture 21

Trefoil v4: Structs

```
(define (interp1 e)
  (cond
    ((= (car e) 'int) (cdr e))
    ((= (car e) 'add)
     (+ (interp1 (cadr e1)) (interp1 (caddr e2))))))
```

## Group Work

Add multiplication to each of these interpreters

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; user-defined helpers for is-int, get-int, make-int, is-add, etc

```
(define (interp2 e)
  (cond
    ((is-int e) (get-int e))
    ((is-add e) (+ (interp2 (get-e1 e)) (interp2 (get-e2 e))))))
```

---

```
(struct int num)
(struct add e1 e2)
; AUTOMATICALLY defines int, int?, int-num, add, add?, add-e1, add-e2
```

```
(define (interp3 e)
  (cond
    ((int? e) (int-num e))
    ((add? e) (+ (interp3 (add-e1 e)) (interp3 (add-e2 e))))))
```